

Multiple Choice (10 marks)

1. What is the density of fat free mass?
 - (a) 0.900 g/cm³
 - (b) 0.730 g/cm³
 - (c) 1.100 g/cm³
 - (d) None of the above
2. Molecules of natural polyunsaturated fatty acids in vegetable oils contain:
 - (a) 18 carbon atoms with one carbon-carbon double bond in the cis configuration
 - (b) 20 carbon atoms with at least two carbon-carbon double bonds in the cis configuration
 - (c) 18 carbon atoms with at least two carbon-carbon double bonds in the trans configuration
 - (d) 18 carbon atoms with at least two carbon-carbon double bonds in the cis configuration
3. The thermic effect of food
 - (a) is substantially higher for carbohydrate than for protein
 - (b) is accompanied by a slight decrease in body core temperature.
 - (c) is partly related to sympathetic activity stimulation in the postprandial phase
 - (d) is not attenuated by food malabsorption.
4. Which of the following does not occur with increasing age?
 - (a) A reduction in lean body mass
 - (b) A reduction in bone density
 - (c) An increased appetite
 - (d) Impaired immune function
5. What are the main enzyme systems for catalysing ethanol to acetaldehyde?
 - (a) Alcohol dehydrogenase (ADH)
 - (b) Catalase
 - (c) Alcohol dehydrogenase (ADH), cytochrome P 450 2 E 1 (CYP 2 E1), catalase
 - (d) cytochrome P 450 2 E 1 (CYP 2 E1)

True / False (10 marks)

Answer True or False

6. True or False: Lactose is excluded on a low FODMAP diet to help manage digestive discomfort.
7. True or False: Macrobiotic and vegan diets are generally low in sugar, focusing on whole, unprocessed foods.
8. True or False: Additional dietary protein may reduce muscle mass and strength in individuals who consume it regularly.
9. True or False: Cholesterol and fatty acids absorbed from the intestinal lumen are packaged into chylomicrons and secreted into the lymph.
10. True or False: Pregnancy and lactation decrease the rate of dietary calcium absorption in the body.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of pepsinogen and gastric lipase in the stomach, including how they are secreted and their functions in the digestive process.
12. Discuss the role of enteropeptidase in the activation of digestive enzymes in the pancreas after a meal. How does this process contribute to digestion?

End of Paper